

Ex. No: 15 Disk Scheduling Algorithms

PROGRAM A: FCFS DISK SCHEDULING

C PROGRAM

```
#include<stdio.h>

#include<stdlib.h>

int main()
{
    int req[20], n, head, i;
    int seek = 0;
    printf("Enter Number of Requests: ");
    scanf("%d",&n);
    printf("Enter Request Queue:\n");
    for(i=0;i<n;i++)
        scanf("%d",&req[i]);
    printf("Enter Initial Head Position: ");
    scanf("%d",&head);
    for(i=0;i<n;i++)
    {
        seek += abs(req[i] - head);
        head = req[i];
    }
    printf("Total Head Movement = %d\n",seek);
    return 0;
}
```

Output:

```

(kali㉿kali)-[~]
$ nano fcfs.c

(kali㉿kali)-[~]
$ gcc fcfs.c -o fcfs

(kali㉿kali)-[~]
$ ./fcfs
Enter Number of Processes: 2
Enter Burst Time for P1: 3
Enter Burst Time for P2: 4

Process BT      WT      TAT
P1      3        0        3
P2      4        3        7

Average Waiting Time = 1.50
Average Turnaround Time = 5.00

```

SHELL SCRIPT

```

#!/bin/bash

queue=(98 183 37 122 14 124 65 67)

head=53

seek=0

for req in "${queue[@]}"
do
    diff=$((req-head))
    if [ $diff -lt 0 ]
    then
        diff=$((
            -diff ))
    fi

    seek=$((seek+diff))

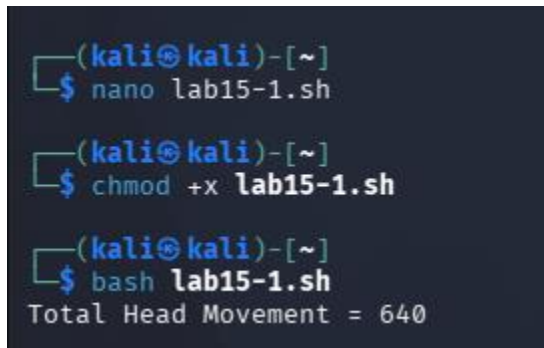
```

head=\$req

done

echo "Total Head Movement = \$seek"

Output:

A terminal window with a dark background and light blue text. The prompt is '(kali㉿kali)-[~]'. The user enters '\$ nano lab15-1.sh'. The prompt changes to '\$'. The user enters '\$ chmod +x lab15-1.sh'. The prompt changes to '\$'. The user enters '\$ bash lab15-1.sh'. The output is 'Total Head Movement = 640'.

```
(kali㉿kali)-[~]  
$ nano lab15-1.sh  
  
(kali㉿kali)-[~]  
$ chmod +x lab15-1.sh  
  
(kali㉿kali)-[~]  
$ bash lab15-1.sh  
Total Head Movement = 640
```

PROGRAM B: SSTF DISK SCHEDULING

C PROGRAM

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
int main() {
```

```
int req[20], visited[20]={0};
```

```
int n, head, i, count=0;
```

```
int seek=0, index, min, distance;
```

```
printf("Enter Number of Requests: ");
```

```
scanf("%d",&n);
```

```
printf("Enter Request Queue:
```

```
\n");
```

```
for(i=0;i<n;i++)
```

```

scanf("%d",&req[i]);
printf("Enter Initial Head Position: ");
scanf("%d",&head);
while(count<n)
{
min=9999;
for(i=0;i<n;i++)
{
if(!visited[i])
{
distance=abs(req[i]
-head);
if(distance<min)
{
min=distance;
index=i;
}
}
seek+=min;
head=req[index];
visited[index]=1;
count++;
}
printf("Total Head Movement = %d\n",seek);
return 0;
}

```

Output:

```
(kali㉿kali)-[~]
$ nano sstfDisk.c

(kali㉿kali)-[~]
$ gcc sstfDisk.c -o sstfDisk

(kali㉿kali)-[~]
$ ./sstfDisk
Enter Number of Requests: 5
Enter Request Queue:
12 45 78 36 23
Enter Initial Head Position: 50
Total Head Movement = 104
```

SHELL SCRIPT

```
#!/bin/bash
```

```
echo "SSTF Disk Scheduling Demonstration"
```

```
echo "Request Queue : 98 183 37 122 14 124 65 67"
```

```
echo "Initial Head Position : 53"
```

```
echo "Total Head Movement calculated using SSTF."
```

Output:

```
(kali㉿kali)-[~]
$ nano lab15-2.sh

(kali㉿kali)-[~]
$ chmod +x lab15-2.sh

(kali㉿kali)-[~]
$ bash lab15-2.sh
SSTF Disk Scheduling Demonstration
Request Queue : 98 183 37 122 14 124 65 67
Initial Head Position : 53
Total Head Movement calculated using SSTF.
```

PROGRAM C: SCAN DISK SCHEDULING

C PROGRAM

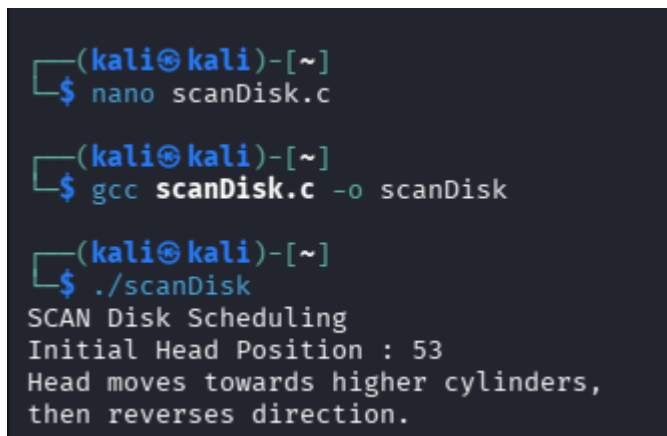
```
#include<stdio.h>

int main()
{
    int disk_size = 200;

    int head = 53;

    printf("SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
    printf("Head moves towards higher cylinders,\n");
    printf("then reverses direction.\n");
    return 0;
}
```

Output:

A terminal window with a dark background and light blue text. The prompt is (kali@kali)-[~]. The user enters 'nano scanDisk.c', then 'gcc scanDisk.c -o scanDisk', and finally './scanDisk'. The output of the program is displayed: 'SCAN Disk Scheduling', 'Initial Head Position : 53', 'Head moves towards higher cylinders,', and 'then reverses direction.'

```
(kali@kali)-[~]
$ nano scanDisk.c

(kali@kali)-[~]
$ gcc scanDisk.c -o scanDisk

(kali@kali)-[~]
$ ./scanDisk
SCAN Disk Scheduling
Initial Head Position : 53
Head moves towards higher cylinders,
then reverses direction.
```

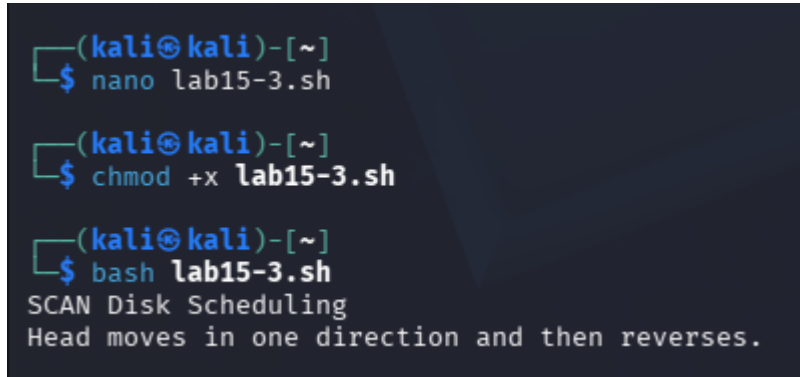
SHELL SCRIPT

```
#!/bin/bash

echo "SCAN Disk Scheduling"
```

echo "Head moves in one direction and then reverses."

Output:

A terminal window on a Kali Linux system. The prompt is (kali㉿kali)-[~]. The user enters 'nano lab15-3.sh', then 'chmod +x lab15-3.sh', and finally 'bash lab15-3.sh'. The output of the script is 'SCAN Disk Scheduling' followed by 'Head moves in one direction and then reverses.'

```
(kali㉿kali)-[~]  
$ nano lab15-3.sh  
  
(kali㉿kali)-[~]  
$ chmod +x lab15-3.sh  
  
(kali㉿kali)-[~]  
$ bash lab15-3.sh  
SCAN Disk Scheduling  
Head moves in one direction and then reverses.
```

PROGRAM D: C-SCAN DISK SCHEDULING

C PROGRAM

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
int head = 53;
```

```
printf("C-SCAN Disk Scheduling\n");
```

```
printf("Initial Head Position : %d\n",head);
```

```
printf("Head moves in one direction.\n");
```

```
printf("After reaching the end, it returns to the beginning.\n");
```

```
return 0;
```

```
}
```

Output:

```
(kali㉿kali)-[~]  
$ nano cdisk.c  
  
(kali㉿kali)-[~]  
$ gcc cdisk.c -o cdisk  
  
(kali㉿kali)-[~]  
$ ./cdisk  
C-SCAN Disk Scheduling  
Initial Head Position : 53  
Head moves in one direction.  
After reaching the end, it returns to the beginning.
```

SHELL SCRIPT

```
#!/bin/bash  
  
echo "C-SCAN Disk Scheduling"  
  
echo "Head moves in one direction only."  
  
echo "After reaching the end, it returns to the beginning."
```

Output:

```
(kali㉿kali)-[~]  
$ nano lab15-4.sh  
  
(kali㉿kali)-[~]  
$ chmod +x lab15-4.sh  
  
(kali㉿kali)-[~]  
$ bash lab15-4.sh  
C-SCAN Disk Scheduling  
Head moves in one direction only.  
After reaching the end, it returns to the beginning.
```